Lecture ①

Rectilinear Motion of one Particle

Basic concepts

① Position (S) or (x) → تحديد مكان جسم معين في الفضاء بالنسبة لنقطة مرجعية ثابتة تسمى نقطة الأصل Origin Point

(كمية متجهة تحدد مكان الجسم بالنسبة لنقطة الأصل)

② Displacement (ΔX) or (ΔS) → المسافة بين الموضع النهائي والموضع الابتدائي للنقطة

$$\Delta X = S_f - S_i$$

③ Distance (d) = $|\Delta X|$ ← في اتجاه واحد

المسافة

$D = d_1 + d_2 = |\Delta X_1| + |\Delta X_2|$ ← في حركة متعددة المراحل أو تغيير الاتجاه

تحسب في الاتجاه

④ Velocity $\frac{dS}{dt}$ → cavg → $\frac{\Delta X}{\Delta t}$ ← السرعة المتوسطة

→ ins → $v = \frac{dx}{dt}$ ← السرعة اللحظية

⑤ Acceleration $\frac{dv}{dt}$

Speed = $|V|$ ← هو مقدار السرعة المتجهة بدون اتجاه

الانطلاقة

(تزداد قيمة الانطلاقة بزيادة مقدار السرعة، بغض النظر عن إشارتها الموجبة أو السالبة)
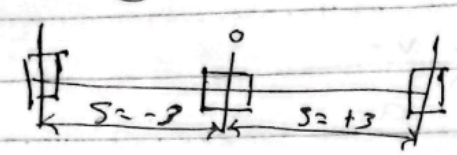
$V_i = +10 \xrightarrow{t_0} V_f = +20$ ← انطلاقة تزايدية

$V_i = +10 \xrightarrow{t_0} V_f = -20$ ← انطلاقة تزايدية أيضاً

معنى أن السرعة هل تزداد أم تقل

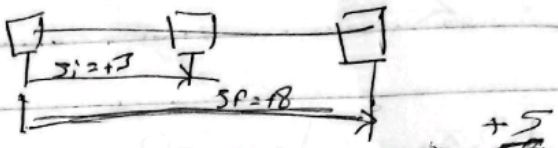
S - Position الموضع — هو مكان الجسم بالنسبة لنقطة أصل ثابتة

Vector متجه ← mag ← ± dir



ΔS - Displacement الإزاحة — هو التغير في موضع الجسم (المسافة المستقيمة من نقطة البداية إلى النهاية)

$\Delta S = S_f - S_i$



$\therefore \Delta S = S_f - S_i = +8 - 3 = +5$

D - Total Distance المسافة الكلية — هي طول المسار الذي يقطعه الجسم كله دائماً magnitude ± only

$D = |\Delta S_1| + |\Delta S_2|$

$D = |+5| + |-5| = 10$

velocity السرعة

AVG السرعة المتوسطة $= \frac{\Delta S}{\Delta t}$

INST السرعة اللحظية $= \frac{dS}{dt}$

magnitude مقدار + direction اتجاه

$V = \frac{dS}{dt}$ ← magnitude ← direction (±)

AVG Speed = المسافة الكلية / الزمن الكلي $= \frac{D}{\Delta t}$

Acceleration معدل تغير السرعة بالنسبة للزمن

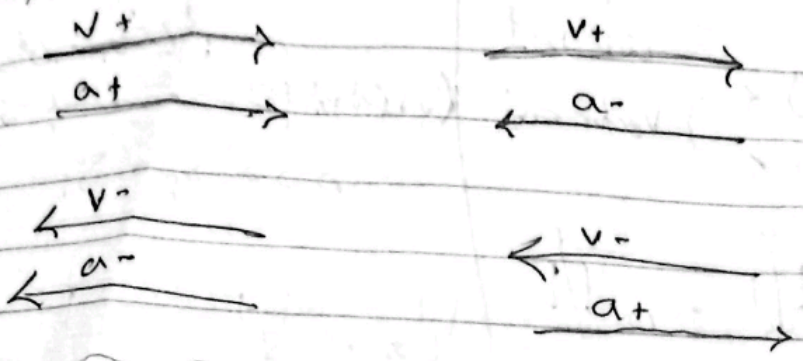
AVG التعجيل المتوسط $= \frac{\Delta V}{\Delta t}$

INST التعجيل اللحظي $= \frac{dV}{dt}$

السرعة كمية متجهة

التباطؤ / التسارع

Acceleration , Deceleration



Example) The Position of a particle along Straight line is given by

$S = (t^3 - 6t^2 - 15t + 40)$ m

where t is second determine

(a) Time when velocity is zero

(b) Position and distance Travelled at that time

(c) acceleration at that time

(d) distance Travelled from $t = 4s$ to $t = 6s$

Solution

(a) $v = \frac{ds}{dt} = 3t^2 - 12t - 15$

$3t^2 - 12t - 15 = 0$

$t = 5$, ~~$t = -1$~~ مرفوض

$\therefore t = 5$ ✓

(b) $S = t^3 - 6t^2 - 15t + 40 = 5^3 - 6*5^2 - 15*5 + 40 = -60$

at $t = 0 \Rightarrow S = 40$ $\Rightarrow$ distance $= |-60 - 40| = 100$

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(c) acceleration at that time

$$a = \frac{dv}{dt}$$

$\therefore a = 6t - 12$

$\therefore a = 6 \times 5 - 12 = 18 m/s^2$

(d) distance travelled from $t = 4s$ to $t = 5s$

$$s = t^3 - 6t^2 - 15t + 40$$

$s(4) = 4^3 - 6 \times 4^2 - 15 \times 4 + 40 = -52$

$s(5) = 5^3 - 6 \times 4^2 - 15 \times 4 + 40 = -60$

$s(6) = 6^3 - 6 \times 6^2 - 15 \times 6 + 40 = -50$

$\Delta s_1 = |-60 + (-52)| = |-8| = 8m$

$\Delta s_2 = |-60 - (-50)| = |-10| = 10m$

$DT = 8 + 10 = 18m$

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lecture ①

Rectilinear Motion of one particle

Example ① Dr/Eslam

The position of a particle on a straight line is given by $s = t^3 - 9t^2 + 15t$ m, where t is in seconds. Determine the position, instantaneous velocity, average velocity, instantaneous acceleration, and average acceleration of the particle when $t = 6$ sec and the total distance it travels during the 6-sec time interval. Also, Plot the Path.

Solution

① Position at $t = 0$, and $t = 6$

$s = t^3 - 9t^2 + 15t \Rightarrow t = 0, \Rightarrow s = 0, \Rightarrow t = 6, \Rightarrow s = -18\ m$

② Displacement ازاحة $\Delta s = s_f - s_i$ المسافة النهائية - المسافة الابتدائية

$\Delta s = -18 - 0 = -18\ m$

③ instantaneous velocity السرعة اللحظية at $t = 6s$

$v_{inst} = \frac{ds}{dt} = 3t^2 - 18t + 15 = 3*6^2 - 18*6 + 15 = 15\ m/s$

$v_{(0)} = 0 - 0 + 15 = 15\ m/s$

④ average velocity السرعة المتوسطة From $t = 0s$ to $t = 6s$

$v_{avg} = \frac{\Delta s}{\Delta t} = \frac{-18 - 0}{6 - 0} = -3\ m/s$

⑤ instantaneous acceleration التسارع اللحظي at $t = 6s$

$a_{inst} = \frac{dv}{dt} = 6t - 18 = 6*6 - 18 = 18\ m/s^2$

⑥ average acceleration التسارع المتوسط From $t = 0s$ to $t = 6s$

$a_{avg} = \frac{\Delta V}{\Delta t} = \frac{V_{(6)} - V_{(0)}}{6 - 0} = \frac{15 - 15}{6} = 0$

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⑦ Total Distance

$$v = \frac{ds}{dt} = 3t^2 - 18t + 15 = 0$$

$\therefore t = 5 \quad , t = 1$

$$t=0 \Rightarrow s(0) = 0 - 9*0^2 + 15*0 = 0$$

$$t=1 \Rightarrow s(1) = 1^3 - 9*1^2 + 15*1 = 7$$

$$t=5 \Rightarrow s(5) = 5^3 - 9*5^2 + 15*5 = -25$$

$$t=6 \Rightarrow s(6) = 6^3 - 9*6^2 + 15*6 = -18$$

$$D_1 = |s(1) - s(0)| = |7 - 0| = 7m$$

$$D_2 = |s(5) - s(1)| = |-25 - 7| = 32m$$

$$D_3 = |s(6) - s(5)| = |-18 + 25| = 7m$$

$$D_{total} = D_1 + D_2 + D_3 = 7 + 32 + 7 = 46$$

24004 - [illegible]

Sheet (1)

[1] The Position of a particle along a straight line is given by the relation $S = (0.3t^5 - 2.7t^2 + 4.5t)$ m where t is in seconds. determine:

(a) The Position of the particle when t = 10s.

solution

$S(10) = 0.3 * 10^3 - 2.7 * 10^2 + 4.5 * 10 = 75$ m

(b) The total distance it trallls during the 10sec time interval

لازم نعرف ما إذا كان الجسم غير حركته أم لا

نساوي السرعة بصفر لنرى إذا توقف في أي زمن

$$v = \frac{ds}{dt} = 0.9t^2 - 5.4t + 4.5 = 0$$

$\therefore t = 5, \quad t = 1$

$\therefore$ الجسم سكن وغير اتجاه حركته مرتين عند $t = 5, t = 1$

$S(0) = 0.3*0 + 2.7*0 + 4.5*0 = 0$

$S(1) = 0.3*1 - 2.7*1 + 4.5*1 = 2.1$ m

$S(5) = 0.3*5^3 - 2.7*5^2 + 4.5*5 = -7.5$ m

$S(10) = 0.3*10^3 - 2.7*10^2 + 4.5*10 = 75$ m

$(0:1)\ \Delta S_1 = |2.1 - 0| = 2.1$ m

$(1:5)\ \Delta S_2 = |-7.5 - 2.1| = |-9.6| = 9.6$ m

$(5:10)\ \Delta S_3 = |75 - (-7.5)| = 82.5$ m

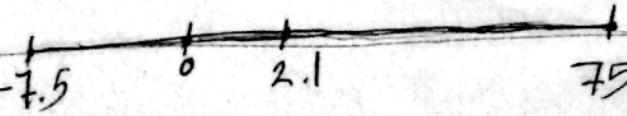
$\therefore D_T = 2.1 + 9.6 + 82.5 = 94.2$ m

(c) The average velocity and average speed during this interval

$$V_{avg} = \frac{\Delta S}{\Delta t} = \frac{S(10) - S(0)}{10 - 0} = \frac{75 - 0}{10 - 0} = 7.5 \text{ m/s}$$

$$\text{AVG speed} = \frac{D}{\Delta t} = \frac{94.2}{10 - 0} = 9.42 \text{ m/s}$$

(d) Plot the Path through this interval of time



[2] The Position of a particle along a Straight line is given by the relation $S = (t^3 - 6t^2 - 15t + 7)$

(a) The total distance it travels during 6 sec interval.

$$V = \frac{ds}{dt} = 3t^2 - 12t - 15 = 0$$

$t = 5$ ~~$t = -1$~~

$S(0) = 7\ m$

$S(5) = -93\ m$

$S(6) = -83\ m$

$(0:5) = |-93 - 7| = |-100| = 100\ m$

$(5:6) = |-83 - (-93)| = |10| = 10\ m$

$D = 100 + 10 = 110\ m$

(b) ~~determine~~ the average velocity and average speed during this interval.

$$N.avg = \frac{\Delta S}{\Delta t} = \frac{S(6) - S(0)}{6 - 0} = \frac{-83 - 7}{6} = -15\ m/s$$

$$\text{Average speed} = \frac{D}{\Delta t} = \frac{110}{6} = 18.33\ m/s$$

1) The Position of a particle → straight line

$$s = t^3 - 6t^2 - 15t + 40$$

(a) The time at $v = 0$

$$v = \frac{ds}{dt} = 3t^2 - 12t - 15 = 0$$

$$\therefore t = 5 \quad , \quad t = -1$$

(b) Position $\Rightarrow s = 5^3 - 6*5^2 - 15*5 + 40 = -60\ m$

distance $\Rightarrow$ from $t = 0$ to $t = 5$

$$s(0) = 40$$

$$\therefore \text{distance} = |-60 - 40| = |-100| = 100\ m$$

(c) $a = \frac{dv}{dt} = 6t - 12 = 6*5 - 12 = 18\ m/s^2$

(d) distance travelled from $t = 4s$ to $t = 6s$

$$s(0) = 40$$

$$s(4) = 4^3 - 6*4^2 - 15*4 + 40 = -52$$

$$s(5) = -60$$

$$s(6) = 6^3 - 6*6^2 - 15*6 + 40 = -50$$

$(4:4)$ $D = |-52 - 40| = |-92| = 92\ m$

$(4:5)$ $D = |-52 - (-60)| = 8\ m$

$(5:6)$ $D = |-50 - (-60)| = 10\ m$

$$D_T = 92 + 8 + 10 = 110\ m$$

$s = t^3 - 9t^2 + 24t - 8$

~~$s = t^3 - 9t^2 + 12t + 10$~~

(a) $v = 0 \Rightarrow v = \frac{ds}{dt}$ $3t^2 - 18t + 24 = 0$

$t = 4s, t = 2s$

(b) $a = 0 \Rightarrow a = \frac{dv}{dt} = 6t - 18 = 0 \Rightarrow 6t = 18 \Rightarrow t = 3$

s at $a = 0 \Rightarrow s = 3^3 - 9 * 3^2 + 24 * 3 - 8 = 10$

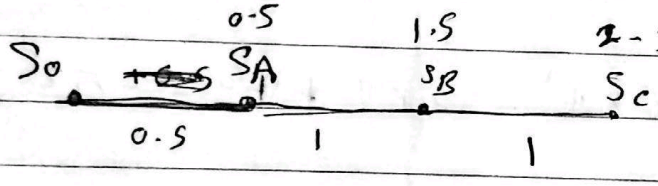
$s(0) = -8$

$s(2) = 12$

$s(3) = 10$

$s(4) = 8$

(7)



$v_{avg} =$

Acceleration → Average Acceleration

$$a_{avg} = \frac{\Delta v}{\Delta t}$$

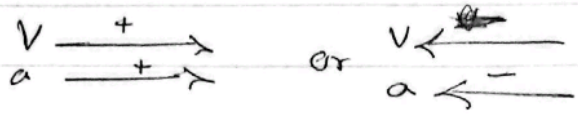
→ Instantaneous Acceleration

$$a_{ins} = \frac{dv}{dt}$$

Acceleration Vs Deceleration

1) تسارع → Speeding up / Acceleration

يحدث عندما يكون السرعة و التسارع نفس الاتجاه / الإشارة



2) تباطؤ Slowing Down / Deceleration

يحدث عندما يكون السرعة و التسارع اتجاهان / إشارتان متعاكستان

